XRPL Validator — Daily Engineering Report · 2026-03-31

Period: 2026-03-31 00:00 to 24:00 US Eastern (2026-03-31 04:00 to 2026-04-01 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 2 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat: resilient RPC client with automatic failover** (`5995a60`, xrpl-validator) Adds RippledClient that tries local rippled first, falls back to xrplcluster.com and s1.ripple.com automatically. WebSocket failover cycles through endpoints on disconnect. Periodically retries primary after 100 successes on fallback.
- **feat: resilient RPC failover + sync log UI + SHAMap test harness** (`95b85f7`, xrpl-validator)
- RippledClient with automatic failover (local → xrplcluster → s1.ripple) - State page now shows live sync log with match/mismatch counters - SHAMap incremental hash tests (10k, 50k, 500k all pass) - Reverted to proven 2.8s flat array for production stability - SHAMap 0.004s approach needs further debugging at 18.7M scale

Independence program

No independence record for UTC day 2026-03-31; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-31 21:49	—	xrpl-validator	`5995a60`	feat: resilient RPC client with automatic failover
04-01 00:09	—	xrpl-validator	`95b85f7`	feat: resilient RPC failover + sync log UI + SHAMap test harness

Times in UTC. A commit appears on the day it was committed to main or merged into it.